

ANGL

$$\Theta = \arctan \left(\frac{x_T - x_s}{y_T - y_s} \right) - v_0$$

$$v_0 = \arctan \left(\frac{x_T - x_s}{y_T - y_s} \right) - \Theta$$

ZEND

$$\cos \varphi = \frac{z_T + H_p - z_s - H_i}{\sqrt{dx^2 + dy^2 + (z_T + H_p - z_s - H_i)^2}}$$

$$\cos^2 \varphi \cdot (dx^2 + dy^2 + dz^2) = dz^2$$

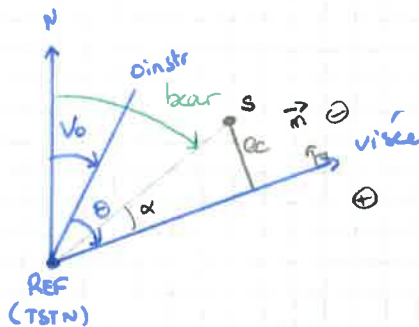
$$\cos^2 \varphi (dx^2 + dy^2) = dz^2 (1 - \cos^2 \varphi)$$

$$\cotan \varphi \cdot \sqrt{dx^2 + dy^2} = z_T + H_p - z_s - H_i$$

$$H_i = z_T + H_p - z_s - \cotan \varphi \cdot d_{2D}$$

ECTH

$$ecth = -\cos(\Theta + v_0)(x_s - x_n) + \sin(\Theta + v_0)(y_s - y_n) - c_s$$



$$bear = v_0 + \Theta - \alpha$$

$$v_0 = \arctan \left(\frac{x_s - x_n}{y_s - y_n} \right) + \arcsin \left(\frac{ec}{d_{2D}} \right) - \Theta$$

DLEV

$$d_3 = (z_n - z_r) + d_{ref} - cd_3 - d_{Tg} \tan \varepsilon$$

$$\varepsilon = \arctan \left(\frac{z_n - z_r + d_{ref} - d_3 - cd_3}{d_{Tg}} \right)$$

DSPT / DIST / DH or

$$\Delta = \sqrt{dx^2 + dy^2 + dz^2} - c_0$$

$$c_0 = \sqrt{dx^2 + dy^2 + dz^2} - \Delta$$

$$(\Delta + c_0)^2 = dx^2 + dy^2 + (z_T + H_T - z_i - H_i)^2$$

$$\sqrt{(\Delta + c_0)^2 - (dx^2 + dy^2)} = z_T + H_T - z_i - H_i$$

$$H_i = z_T + H_T - z_i - \sqrt{(\Delta + c_0)^2 - (dx^2 + dy^2)}$$

PLR3D

$$(\Delta + c_0) \begin{pmatrix} \sin \theta \sin \varphi \\ \cos \theta \sin \varphi \\ \cos \varphi \end{pmatrix} - A \begin{pmatrix} x_T - x_s \\ y_T - y_s \\ z_T + H_T - z_s - H_i \end{pmatrix} = 0$$

Pour c_0 on peut utiliser n'importe quelle ligne.

$$(\Delta + c_0) \cos \varphi = a_{31} dx + a_{32} dy + a_{33} dz$$

$$c_0 = \frac{a_{31} dx + a_{32} dy + a_{33} dz - \Delta \cos \varphi}{\cos \varphi}$$

si $\cos \varphi \neq 0$.

si $\cos \varphi = 0$ utiliser ligne 2 si $\cos \theta \neq 0$ sinon ligne 1.

$$* V_0 \rightarrow R_x = R_y = 0 \text{ \& } H_i = 0$$

$$(1) (\Delta + c_0) \sin \varphi \sin \theta - \cos V_0 dx + \sin V_0 dy = 0$$

$$(2) (\Delta + c_0) \cos \theta \sin \varphi - \sin V_0 dx - \cos V_0 dy = 0$$

On résout l'équation n°2.

$$A \cos X + B \sin X + C = 0.$$

(cf ECSP)

$$* R_x \quad \begin{cases} V_0 = R_z = 0 \\ H_i = 0 \end{cases}$$

Attention $V_0' = V_0 \rightarrow$ acst en fait

$$(1) (\sigma + \cos) \sin \theta \sin \varphi - \cos(-\alpha) dx + \sin(-\alpha) \cos R_x dY - \sin(-\alpha) \sin R_x dz$$

$$(2) (\sigma + \cos) \cos \theta \sin \varphi - \sin(-\alpha) dx - \cos(-\alpha) \cos R_x dY + \cos(-\alpha) \sin R_x dz = 0$$

$$(3) (\sigma + \cos) \cos \varphi - \sin R_x dY - \cos R_x dz = 0$$

On résout (3) pour trouver R_x .

$$* R_y \quad \begin{cases} V_0 = R_x = 0 \\ H_i = 0 \\ \text{ACST } \triangle \end{cases}$$

$$(1) (\sigma + \cos) \sin \theta \sin \varphi - \cos(-\alpha) \cos R_y dx + \sin(-\alpha) dY - \sin R_y \cos(-\alpha) dz = 0$$

$$(2) (\sigma + \cos) \cos \theta \sin \varphi - \sin(-\alpha) \cos R_y dx - \cos(-\alpha) dY - \sin(-\alpha) \sin R_y dz = 0$$

$$(3) (\sigma + \cos) \cos \varphi - \sin R_y dx - \cos R_y dz = 0$$

On résout l'équation n° 3.

$\vec{EWP}$

ECSP

$$ecsp = \sqrt{d^2 - (dx \cos \theta \sin \varphi + dy \cos \theta \sin \varphi + dz \cos \varphi)^2} - d \cos \varphi$$

II/ faut résoudre

$$A \cos X + B \sin X + C = 0$$

$$A = dx \sin \varphi$$

$$B = dy \sin \varphi$$

$$C = \sqrt{d^2 - (ecsp + d \cos \varphi)^2} - dz \cos \varphi$$

On fait le chgt de variable

$$t = \tan\left(\frac{X}{2}\right) \text{ avec } X = \theta + \varphi_0$$

$$\sin X = \frac{2t}{1+t^2} \text{ et } \cos X = \frac{1-t^2}{1+t^2}$$

→ eq. 2nd degré

$$at^2 + bt + c = 0.$$

avec

$$\begin{aligned} a &= C - A \\ b &= 2B \\ c &= C + A \end{aligned}$$

$$V_0 = 2 \cdot a \tan(t)$$